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FEATURES

Racial profiling for induction of labour: improving safety or perpetuating racism?

julietwalker · The BMJ Blog

The disproportionate mortality and morbidity in mothers and babies from Black, Asian, and ethnic minority backgrounds as compared to white mothers and babies is clearly evidenced. [1-3] There have been efforts at a national level to tackle the issue. For example, the Royal College of Obstetricians and Gynaecologists (RCOG) held a public engagement event and developed partnerships with maternity user groups. [4,5] However, in an attempt to lessen this disproportionate morbidity and mortality, the National Institute for Health and Care (NICE) has, within its new draft guidelines for induction of labour, a suggestion of racial profiling. They recommended that women from ethnic minority backgrounds should consider having their pregnancy induced at 39 weeks, even if the pregnancy has no complications. [6] The recommendations have led to significant backlash from advocacy groups and doctors. [7-9]

Historic, systemic biases in medicine have arisen from biological determinism. Cerdeña et al's paper on race-based medicine pointed to its inherent biases that are pernicious and ongoing within clinical medicine. [10] Proponents of critical race theory argue that race is a social and political construct where "bodies inherit not merely genes, but power relationships, legacies of discrimination, the ideological effects of past social policy, and generational systems of belief." [11] Never the less, intersectional oppressions can lead to the epigenetic phenomena of weathering where hardships can produce disease pathology. [12] Thus, making recommendations based on race alone must be critically evaluated and never undertaken in isolation.

We are deeply concerned that if these recommendations are taken forward uncritically, they could further embed institutional racism in maternity care, strengthen racial biases and stereotypes, legitimise skin tone as clinically meaningful, pathologize healthy pregnancies in women from ethnic minority backgrounds, and undermine choice for black and brown women.

We want to draw attention to the concept of Cultural Safety in which structural reflexivity is more important than reflectivity [13]. "Reflectivity" involves analysing what has happened. However, "reflexivity" involves self or institutional assessment, evaluation of power imbalances, and reaction to the circumstances as they are happening. Its purpose is to look inwardly and outwardly in a social context. This would lead to critically appraising the evidence base for structural/institutional racism; to acknowledge race as a social construct and racism as a determinant of health; and recognition of the lived experiences of women from ethnic minorities and birthing people within healthcare systems, and co-production as essential in generating guidelines.

In the NICE evidence review for the draft guidelines the lack of direct evidence for women from ethnic minorities is noted by NICE. Of the studies referenced the vast majority did not record race or were unable to, or failed to, report on ethnic variation due to low numbers of minority ethnic women. [14] This "absence" of evidence could be construed as a form of structural racism.

Attention has been drawn to older studies from routine data sets that suggest different gestation lengths for women from ethnic minorities in comparison to white women. [15-17] In critically appraising this further, these routine data are of a lower quality and not from controlled trials. Studies from older routine health data can fall prey to bias, and indeed mask or conceal structural discrimination and racism, and should only be used as a signal for launching better studies. Indeed an examination of a data set of birth outcomes for African and Caribbean babies in England and Wales makes a case against over homogenisation of women from ethnic minorities. [18]

In a statement from the RCOG about NICE's draft guidance the college imply that induction has no downsides, but they don't seem to have taken into account the recent long term adverse outcomes data for inductions of labour in uncomplicated pregnancies from Australia, or the increasing evidence that the risk of stillbirth is reduced by amplifying continuity of midwifery care models. [19-22]

Achieving high quality national guidance also requires an examination of the impact of social, cultural, and political systems on health, wellbeing, safety, access to care, quality of care, and autonomy. Structural racism is pervasive across British society. Racism is a known determinant of health, occurring at systemic and individual levels. Its role in perpetuating the extreme disparities witnessed in maternity care needs to be addressed through "race conscious medicine" as described by Cerdeña et al. [10]

Experiential data can enhance critical analysis by positively challenging biases and reductive stereotyping and exposing racism that quantitative data may conceal. There are widespread qualitative data which show that women in all ethnic minority groups have poorer experiences of care across antenatal, intrapartum, and postnatal stages than white women. [23] Bringing together experiential knowledge and priorities with clinical knowledge and priorities in co-production processes would increase not only the quality of guidance, but increase confidence in it, and help achieve individualised care for all women, and reduce coercion. While NICE have been emphatic that guidelines should not be tramlines, efforts to address the lack of birthing women's autonomy in induction of labour requires significantly more attention. [24,25]

The conversations around the NICE draft guidelines provide strong justification for structural reflexivity, "race conscious" medicine, and co-production. [10,13] The opportunities for real change should not be overlooked. If indeed future higher quality studies reveal different gestation lengths for women and birthing people from ethnic minorities, there should be no delay in presenting these data to stakeholders.

Joining Stripe

James Long (jlongster)

I made a big change recently: I joined Stripe as a developer on a team that works on the dashboard. Today is my first day!

First, let me say that I'm really excited. I talked to Alex Sexton 5 years ago about joining Stripe, but the timing wasn't quite right. It's always been on my list of companies (which is quite small) that I would work for if I was to make a change. The problem is interesting, lots of great people work there, and they are remote friendly.

This comes with a tinge of sadness, though. Over 3 years ago I quit my job to explore contracting and focus more on building a product. That product eventually become Actual , and I focused hard on it for the last year and a half. "Getting a job" feels like quitting.

I've struggled to identify my goals. Am I a startup? Should I get funding? Do I need to find a cofounder? What happens in 5 years?

The truth is: I love building products. I love talking with customers and solving their needs. I love figuring out how to market it. But when it comes down to it, building a business is not something I want to do. There's so much work involved and it's just too stressful for my family.

I had a goal of reaching 1000 users which would start to be financially sustainable. It seems like a modest goal, but I've learned how hard it is to actually sell software. I've only reached 155 users, and there's tons of reasons why I haven't grown faster. The biggest problem is trying to do this alone — besides amplyifying my weaknesses, there's simply too much work to do.

I didn't want to do it alone, but I failed at finding a cofounder or somebody who would be significantly involved without funding. I thought about getting funded, and even talked to a few investors, but it exposed my feelings deep down that I didn't really want to be in the business of building a business.

I just want to make a great product that users love. And I'm going to keep doing that.

What does this mean for Actual?

In short: Actual is still very much alive, but I won't have as much time to work on it. Let me try to explain why that might be a good thing.

Actual is a personal finance tool focusing on providing the most power and flexibility in the simplest tool possible. The simplicity is expressed in a fast and streamlined UX, and the goal is to provide powerful tools for managing your finances how you like (choose your budgeting method, write custom reports, and more).

Actual is the hardest thing I've ever done. I like to rethink things from the bottom up, and Actual is no exception. Inside of Actual is a syncing engine that keeps all of your data local and uses smart techniques for robustly syncing changes across devices. There's so much cool stuff in there, like end-to-end encrypting the changes so you have full privacy.

The entire app is what's called "local-first": literally everything is happening against local data, and syncing to the cloud is just a background service that happens when it can. This changes everything.

One of the biggest advantages is the app is super fast . Since everything is local, reading and writing data is stupidly fast. You can try the app for yourself here .

It's all custom built because nothing out there met all my requirements. And I don't regret building it at all. I'm confident that I wouldn't be able to do everything I've done with anything else.

Just a sampling: full undo/redo (like, the real thing, not some cheap version), running the app in the background that multiple tabs/windows connect to, using the app with literally nothing going over the network , and lots more.

Note how all of these benefits are user-facing . I wouldn't invest so much in this if I didn't think it made a huge impact on the capabilities of the product itself. A driving motivation for this architecture is the ability to write custom reports in a simple language, and it'll be easy since it has access to all your data right there.

The thing I'm most proud of is that the syncing engine has reached almost 100% stability. Any problems are now caused by external factors, like a bug in the code that initially downloads your data when logging in. The layer that actually commits changes and syncs them across devices is nearly 100% reliable: if you make a change, sync to the cloud, and sync other devices, you are guaranteed to see the same data. And it's all totally seamless.

Unfortunately, it's costly to build things up from scratch. Here's what surprised me: writing the code itself didn't take too long. However, when you've done something different, it takes a lot of work to explain it to users and build up a coherent UX.

For example, now that everything is local, what does the login process look like? Do I authenticate a user, and then get them to manually copy over data from another device to get started? Well... of course not. I built a system for tracking which files are available to download, but what happens once you're all set up if you use "reset sync" on one device? "Reset sync" deletes all syncing info and starts fresh, but now other devices won't sync and need to be re-setup.

It sure would be nice if I didn't have to worry about all this, but moving fully to the cloud just gets rid of too many nice features. I had to rebuild the whole user experience for

The Skill of Using AI Agents Well

becausecurious · LessWrong

AI usage for this post: I wrote the draft on my own. While writing, I used Claude Code to look up references. Then Claude Code fixed typos and reviewed the draft, I addressed comments manually.

Epistemics: my own observations often inspired by conversations on X and Zvi's summaries.

As Zvi likes to repeat Language Models Offer Mundane Utility . Agent harnesses is the most advanced way to use language models. At the same time, they are not perfect - the capabilities frontier is jagged, sometimes they make mistakes and sometimes they just nuke your 15 year photo collection or production database . Thus, it is a skill how to use AI agents efficiently and I want to get better at it.

What tips, tricks, approaches are you using to improve your efficiency when using agent harnesses? I personally focus on Claude Code & Codex CLI mostly for single person software development, but I welcome suggestions for other tools and other areas of use. Ideally you share what you tried and what works for you and I try it myself and see whether it improves my workflow.

Here are my discoveries (with level of confidence).

Use best model and highest thinking/effort (no brainer)

The best available model and its thinking effort usually produces best results and requires less handholding. You have to be extra careful with what this means, e.g. Claude Code has high thinking effort by default, but the best level is max, which you need to actively turn on yourself. Unless you are very sensitive to speed and cost, you should do this. It might work a bit longer and consume more tokens, but this is much better than requiring multiple iterations from you.

worktrees (no brainer)

When working on code in a git repo, let each AI session have its own git branch with worktree. This way they can work in parallel and not fight to edit same files. Intuitively this would lead to merge hell, but fortunately AIs are good at merging.

/fast (depends on how you work)

Codex CLI allows you to turn on /fast mode, which speeds up processing 1.5 times, but consumes your token quota 2x faster. If you work on highly interactive tasks, which require lots of your input and you are not constrained by money (e.g. can afford $200 subscription), you should do this. In my experience, without /fast, AI is slow enough that I can juggle 5-6 sessions before any finishes. With /fast, sessions finish quickly enough that I only need 2-3 at a time. The smaller batch means less context switching, which I find more productive overall.

The toggle is global, so it affects all your sessions at once.

Claude Code also has /fast to speed up processing 2.5 times, but don't run to turn it on yet - it charges you extra in addition to subscription using API prices and it is very expensive ($2-5/minute/agent). I haven't tried it.

Verbose logging (depends on what you are working on)

I discovered that my software development efficiency grew greatly once I started having extremely verbose logging. The idea is once something goes wrong, this gives AI a trace of what has happened and I can just describe very briefly the higher-level symptom and it can investigate from the logs on its own. This is especially useful when the issue is hard to reproduce or happens occasionally. Another variation of this is to have some way to export debug information from your app and reference a specific object or instance. This way when something goes wrong, you just feed that to AI after one click and let it investigate.

AIs have bad days (sometimes)

I don't know the reason, but once in a while I get a couple days when AI just seems incapable of doing anything. Previously after one message it would correctly implement a bunch of stuff and now it keeps misunderstanding what I want and I need 30 iterations and it still does not get it and the process never converges. In this case I switch to the competitor (basically Opus 4.6 <> GPT5.4). The key is to detect this early and switch early. This is very sad, because migrating all the skills and setups between Claude Code and Codex sucks and I don't have an efficient way to do this.

Memory across sessions (WIP)

I haven't found a solution to this myself, but I strongly believe that this would have huge impact. I want the current AI session to seamlessly have access to all past information I provided to it. Both Claude and ChatGPT have this implemented in their web interfaces, but not in Claude Code / Codex CLI. I am currently experimenting with github.com/doobidoo/mcp-memory-service . Issues discovered:

It is developer oriented (e.g. mostly technical facts and choices), but I want generic memory system (e.g. for personal background too).

> *You are the bottleneck This is a guiding principle for multiple ideas.*
>
> becausecurious

Autosaving of memories does not work - Claude ignores CLAUDE.md and the session end approach seems to be regexp based.

The memory retrieval seems to be fine. I am experimenting with using a separate Claude session in the background to extract memories from every message.

Government indecision is still costing lives in the UK

kellybrendel · The BMJ Blog

The government must act now, or be faced with much tougher decisions and less popular choices as the winter kicks in, says John Middleton

One person in 55 is now infected with coronavirus in the UK . The odds are that every time you step onto a crowded bus or train carriage you meet someone who is infected with covid-19. Every time you are sitting in a restaurant maybe, or walking down a supermarket aisle, you will likely meet someone who is infected too. In early October one in 12 children were estimated to have had covid, and on 22 October there were 180 deaths in the UK. Every four hours, as many deaths are occurring in the UK as New Zealand has had for the whole pandemic.

We are still in groundhog day , with a few new repeating scenes. The secretary of state for health and social care, Sajid Javid, has repeated what he said when first taking up the role: that “ there could be 100 000 cases a day ,” yet he doesn’t see this as a need to act, either then or now. English directors of public healt h have again broken from national guidance that they know is inadequate for their areas. NHS leaders and the BMA have called for urgent action “to protect the NHS.”

Plan A of the government’s winter plan has only ever been the “do as little as possible” option. The government’s eggs are all in the vaccination basket: offer flu vaccination and a third dose of the covid vaccine as a booster (or not) then let everything else run loose and it’s the people’s fault if they get ill or infect others.

Malta, Spain, Portugal, and the Netherlands have now overtaken the UK and Israel in the vaccination stakes . France, which had a high level of vaccine hesitancy, has climbed up to a comparable level of coverage to the UK, mainly because the French are using vaccine passports. Eastern and southern Europe, along with the Baltic countries , are now experiencing severe epidemics. They have had high levels of vaccine hesitancy and lower vaccine uptake. In some cases, health systems have been less well prepared to implement mass vaccination, or lost out in the chaotic EU central purchase earlier this year.

Vaccination is still the best hope we have at reducing the virus to very low levels. Countries that have the highest rates of vaccination have seen an encouraging levelling of death rates throughout this year. Yet there is still no evidence of collective (“herd”) immunity even in countries where there is a very high level of vaccination or where there have been persistent waves of infection, as in Iran. Reinfection occurs, especially with the delta variant , and vaccine efficacy declines over time . Vaccines still have a high level of efficacy against the delta variant, although less than with early variants. Yet where there are areas of low vaccine uptake within countries and with social mixing, these weaken a country’s ability to suppress the virus. In the UK there are areas where public confidence in the vaccine is being improved with interventions from local public health teams and community support, but more investment is needed in this localised action. Vaccines prevent serious illness as measured by hospital admission and death but do they prevent debilitating illness—persistent or long covid ? Social mixing without masking enables the virus to find new unprotected, vulnerable people to infect and harm. Every time the virus finds a new host, new mutations occur, and new, potentially more vaccine resistant, variants will be formed in the UK’s “new variant factory. ” So vaccination is not the sole answer.

UK politicians have been too attracted to single technological fixes (testing, vaccination, the covid app, Nightingale hospitals), when what we need is the “Swiss cheese model” —using everything we’ve got at the same time to prevent viral spread. All of the European countries now enjoying high levels of double vaccine coverage with MRNA vaccines have still kept their social measures in place, in differing degrees throughout the summer. Masks have continued to be a requirement in public enclosed spaces in France, Portugal, and Spain. Germany requires FFP2 masks in enclosed public spaces and vaccine passports. Gatherings in public places have been strictly limited in Portugal and Spain until recently, despite high levels of vaccine coverage. France, Ireland, Montenegro, and Israel are other examples of countries requiring vaccine passports.

Living with covid does mean masking up in enclosed spaces, meeting outdoors when possible, working from home if we can, and if we can’t do without our night time entertainment it means vaccine passports. Vaccination, or evidence of a recent negative test, are a small price to pay for “normality.” Every vaccinated person has their vaccination card, and at least 10 million people now have the NHS app. It would seem a very small step to have to show it to gain access to somewhere you want to go—especially where employers should be protecting their staff. Millions of airline passengers have accepted the testing regimes of countries they wish to go to and the mask requirement on planes. The requirements in the UK plan B need to be put back into law for consistent control of the virus across the UK. None of them are major restrictions on our freedom.

The government has now asked local authorities what they think about moving towards plan B . They are, once again, caught stalling on actions that experts called for in the summer. Government ministers trivialise what is at stake with the ill informed debate about masking in Parliament, with careless contributions from the leader of the House of Commons , the care minister , and now the chancellor . They have chosen to pick a destructive and meaningless fight over the non-issue of face to face consultations in primary care, when they should have been praising the extraordinary effort of healthcare staff who’ve rolled out five rounds of covid and flu vaccinations. High levels of satisfaction with GP services continue , patients who need to get a face to face consultation get them, and the digital revolution envisaged by the NHS Long Term Plan has arrived.

Academic medicine and publishing from developing countries

kellybrendel · The BMJ Blog

Samiran Nundy, Atul Kakar, and Zulfi Bhutta have published a book titled How to Practice Academic Medicine and Publish from Developing Countries? A Practical Guide . It's a book that will be extremely useful to the growing number of academics working in low and middle income countries. The book is published by Springer and will from 30 October be available open access, meaning you can access it for free as often as you want. Hard copies will also be available for a fee. I felt privileged to be asked to write the foreword, and what follows is an edited version of my foreword.

When I became an assistant editor at The BMJ in 1979 and began to process scientific papers submitted to the journal it was extremely unusual to see, let alone publish, a paper from a low income country. I remember being surprised by high quality papers coming from Bangladesh, a country that Henry Kissinger called a "basket case." Those papers came from the International Centre for Diarrhoeal Disease Research, Bangladesh, which I learnt later was a creature of the Cold War and might cruelly be called a "branch office of Johns Hopkins."

Years later, in 2013, I became the chair of the board of what was by then called icddr,b. The centre has now published major vaccine trials led by Bangladeshi scientists in the New England Journal of Medicine , 1 and I was privileged to be on the steering committee of a major trial, also published in the New England Journal of Medicine , of a system for managing hypertension in rural Pakistan, Bangladesh, and Sri Lanka led by scientists from those countries. 2 Last year I was delighted to see a major trial of the polypill for the prevention of cardiovascular disease led from India and also published in the New England Journal of Medicine . 3

High quality research relevant to the needs of low and middle income countries is much commoner now than it was 40 years ago, and China has become a scientific leader. But, as the editors of this book describe in the introduction, there is still not nearly as much good research as there should be from the part of the world that carries most of the disease burden. Even worse, the medicine practised in some of these countries is disconnected from research and teaching, and driven more by profit than what is best for patients and the population. I have Indian friends who are terrified of seeing a cardiologist for fear that they will be given treatments they don't need.

I agree with the editors' diagnosis that "the main reasons for our having sunk into this deep morass is not because we are poor but because we have not intelligently examined, evaluated and investigated how we could use our own resources more effectively. We have tended to blindly follow what is being done in richer countries instead of trying to provide healthcare to our population which is accessible, affordable and, most importantly, appropriate even if this means deploying and working with informal healthcare providers."

Other people's research can be valuable, but it can never be as valuable as your own—addressing the problems that matter to your people with relevant methods and the tools you have. And we know that the very act of researching brings improvement, and (as I know to my cost) you can never learn about research from reading about it: you need to do it.

I've never quite understood why people in low and middle income countries would want to replicate the health systems of high income countries. Not only are those systems not relevant to the needs and circumstances of the low and middle income countries, but the systems in high income countries are increasingly unaffordable and unsustainable and not meeting the needs of their own populations.

Health systems in high income countries were developed decades ago and were designed to respond to the infectious disease and trauma that were then the main causes of suffering and death. Those problems could be cured, but now non-communicable disease is the main cause of suffering and death. Such disease cannot be cured and needs a different approach.

Non-communicable disease is now also the main cause of suffering and death in low and middle income countries (apart from some sub-Saharan countries, but even there it will soon be the main cause). The epidemiological transition happened very fast in low and middle income countries: in Bangladesh non-communicable disease caused about 10% of deaths in 1986 but nearer 80% by 2006. 4 I spent years working with 11 centres in low and middle income countries that were doing research, building capacity, and advising on policy in relation to non-communicable disease. We envisioned what a better system in low and middle income countries might look like—with an emphasis on public health, the social determinants of health, prevention, primary care, patient empowerment, and widespread use of evidence based guidelines. 5 (Such guidelines were developed by academics in South Africa as part of a package that allows good primary care where doctors are few or unavailable. 6)

We should have said more about the use of technology. Most people in low and middle income countries, even some of the poorest, now have mobile phones, which has meant that people can communicate without having to connect every house by wires, as happened with terrestrial phone systems in high income countries. Low and middle income countries can in this way "leapfrog" over a stage that was needed in high income countries, and the same can be done for health—not least by using mobile phones to provide access to care. Similarly, health systems in low and middle income countries might create health record systems where patients, not healthcare providers, own and control the records. Health systems in high income countries are just beginning to recognise the importance and inevitability of giving patients ownership and control of their records. (I have a conflict of interest here as I'm the chair of Patients Know Best, a company that gives patients in Britain and some other countries control of their records

Achieving long-term retention of GPs will require funding, support, and a reduction in workload demands

julietwalker · The BMJ Blog

The shortage of general practitioners in the NHS is widely acknowledged , but this has not resulted in the kinds of changes that will make a difference. Last week, Martin Marshall, c hair of the Royal College of General Practitioners, spoke out about the current crisis in general practice . He highlighted how the shortage of GPs, coupled with increased patient demand, is leaving many GPs feeling that they cannot provide a safe level of care. As a result, many GPs are leaving the profession as a consequence of burnout, or are taking early retirement.

As part of our work at the Policy Research Unit in Behavioural Science , (a member of a National Institute for Health Research programme to deliver policy related research), we summarised the evidence about what motivates GPs to take early retirement and reduces occupational participation . We considered the strategies that may prevent withdrawal from the work force and help support increased GP recruitment especially in rural areas. We were also interested in the evidence of effectiveness for behavioural insights, aka nudge-style interventions.

The findings of our report were clear . GP work is increasingly stressful, forcing many to opt for early retirement. The financial arrangements allow many to do so. Golden handshakes alone won't fix rural recruitment. And behavioural insight interventions are highly unlikely to play any effective role in keeping GPs in the profession.

The determinants of stress are numerous: excessive workload, fear of litigation, the administrative and emotional burden of medical revalidation, job dissatisfaction, poor work-life balance, and pessimism about the future of the profession. Qualitative research revealed a number of factors which all contribute to feelings of stress and anxiety among GPs—the emotional toll of managing patients' psychosocial needs; abusive or confrontational patients; a practice culture characterised by conflict or bullying; working in isolation without support; work role demands, specifically a fear of making mistakes; managing patient complaints, appraisal, revalidation, CQC inspections, and financial pressures faced by partners .

Recruitment and retention in rural areas is complex . Multidimensional approaches may be more successful than those relying on financial incentives alone, as lifestyle or personal values are highly influential. Locations of family, partner or spouse were factors that were prioritised over financial incentives to accept a rural post, as well as the ability to control working hours, professional development, a preference for larger practices, paid holiday, and assistance with partners employment and childcare .

Behavioural insights tend to focus on automatic, often unconscious decisions. Instead of supporting individuals to make decisions where they will weigh-up costs and benefits, behavioural insights focus on configuring options to account for human fast thinking . For example advocating a policy intervention that makes changes to the context in which individuals make decisions, rather than attempting to change how individuals feel about/react to contexts. To this extent there are limited opportunities to reduce early retirement, as the choice to retire is conscious and not passive. This was backed-up by the limited number of studies and the modest evidence of effect. However, organisational initiatives that might help physicians deal with stress , include coping strategies, and reflective groups, in addition to configuring work to enable part-time and/or flexible working.

Will the covid-19 pandemic be the catalyst to deliver the seismic change to GP working practices? The additional pressures created by the pandemic, as well as changes to pensions, and a lack of funding, are all counterproductive to long term retention. GP leaders are also unimpressed with their perceived lack of support for clinical practice .

GPs want to work and they are vital. Their compassion and dedication has shone through in the response to covid-19. But achieving long-term retention, and reversing the decline in GPs' morale, will take more than a nudge . It will require funding for more doctors, nurses, and support staff, pension reform, a reduction in workload demands, and additional mental health support. The will of GPs is clearly present, but the environmental conditions are missing.

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How one word in PostgreSQL unlocked a 9x performance improvement

James Long (jlongster)

At the very heart of Actual is a custom syncing engine. Recently I implemented full end-to-end encryption (not released yet) and it inspired me to audit the performance of the whole process. In the future I'll blog more about using CRDTs for syncing, but for now I'd like to talk about a PostgreSQL feature that enabled a 9-10x performance improvement.

Actual is completely a local app and syncing happens in the background (using CRDTs). This means the server is very simple and all it has to do is store and fetch "messages" for clients. The entire code for handling syncing is only 200 lines of JavaScript.

We need to handle a lot of messages to keep syncing fast. In fact, while working on this something strange happened: a new user generated 169,000 messages on one day. This is an outlier by a huge margin. For example, importing 1000 transactions into the system would generate about 6000 messages and, while reasonable, is still more than the average number of message per day per user. I believe they did this by using the API trying to bulk import a lot of data and we have different APIs for that. Still, I thought, what if I made 169,000 my benchmark?

I tried pumping 169,000 messages through the system and broke the server. The request timed out and the server was still crunching through messages making everything else slow. I knew what the problem was instantly.

Messages are stored in PostgreSQL and the table looks like this:

CREATE TABLE messages_binary (timestamp TEXT, group_id TEXT, is_encrypted BOOLEAN, content bytea, PRIMARY KEY(timestamp, group_id));

It stores small binary blobs marked with a timestamp and a "sync group" they belong to.

The server was choking trying to insert so many rows. Unfortunately, we can't simply execute one query with a bunch of INSERT statements when adding messages. Our CRDTs have a few constraints:

Message can't ever be duplicated (identified by timestamp)

We need to update a merkle trie depending on whether or not the message was added

Solving #1 is easy. Because we made timestamp the primary key, we can do INSERT INTO messages_binary (...) VALUES (...) ON CONFLICT DO NOTHING . The ON CONFLICT clause tells it to do nothing when there's a conflict, and duplicates conflict on primary key.

A much bigger problem is #2 . We need the result of the insert to know if a row was inserted or not. If it was inserted, we need to also update our merkle trie like this:

if (inserted) { trie = merkle.insert(trie, Timestamp.parse(msg.timestamp)); }

It's extremely important that each timestamp in the system only ever get inserted to the merkle trie once. The trie is responsible for guaranteeing consistency in the system and maintains hashes for the content. If you haven't added each timestamp once and only once, the hashes (and thus verification) are wrong.

The whole code for updating the database looks like this (using some abstractions over node-postgres):

await runQuery('BEGIN'); let trie = await getMerkle(runQuery, groupId);

for (let message of messages) { let timestamp = message.getTimestamp(); let isEncrypted = message.getIsencrypted(); let content = message.getContent();

let { changes } = await runQuery(`INSERT INTO messages_binary (timestamp, group_id, is_encrypted, content) VALUES ($1, $2, $3, $4) ON CONFLICT DO NOTHING` , [timestamp, groupId, isEncrypted, content]);

if (changes === 1) { /\/ Update the merkle trie trie = merkle.insert(trie, Timestamp.parse(timestamp)); } }

await runQuery(`INSERT INTO messages_merkles (group_id, merkle) VALUES ($1, $2) ON CONFLICT (group_id) DO UPDATE SET merkle = $2` , [groupId, JSON .stringify(trie)]);

await runQuery('COMMIT');

This is mostly the real code, the only difference is we also rollback the transaction on failure. It's extremely important that this happens in a transaction and both the messages and merkle trie are updated atomically . Again, the merkle trie verifies the messages content and they must always be in sync. The user will see sync errors if they are not.

The problem is immediately clear: we are executing an INSERT query for each message individually. In our extreme case we are trying to execute 169,000 statements. PostgreSQL lives on a different server (but close) and making that many network calls alone is going to kill performance, not to mention PG overhead.

I knew this was slow, but I didn't realize how slow. Let's test a more reasonable number of messages that actually finishes. 4000 messages takes 6.9s to complete . This is just profiling the above code, and not taking into account network transfer.

This is a huge UX issue. While this is processing the user is sitting there watching the "sync" icon spin and spin and

A lifting week

Petr Mitrichev · Petr Mitrichev (competitive programming)

The Universal Cup held its Stage 2: Paris during the Oct 6 - Oct 12 week (problems , results , top 5 on the left). This was one of the very rare occassions where the winner seemingly was not a veteran team, but a team THU1 consisting of university students. They were already ahead of the usual suspects on 12 problems by penalty time, and solving the last problem 10 minutes before the end was the icing on the cake. Congratulations!

Codeforces Round 1058 wrapped up the week on Sunday (problems , results , top 5 on the left, analysis , my screencast). As the custom goes these days, I have started well on the easier problems but then got stuck, trying to make progress in either of the three remaining problems, and actually being close to solving D2 at the end, but not quite.

Nobody was able to solve everything this time, but the top 5 shows that many different strategies were competitive. In the end solving the problems fast from left to right was the winning recipe for Benq. Congratulations!

In my previous summary , I have mentioned another Codeforces problem : you are given n a l + z , and then the first chain will have a l +1 at the second position. However, because the jump function is monotonic, the second chain will always be at the same position or to the right of it for the same step, and the first chain will always be at the same position or to the right of it when one step ahead of the second chain. So we just need to check those two possibilities (a collision at the same step, or a collision where the first chain is one step ahead) in the binary lifting condition.

This way we can precompute for all n options for the starting position l , where, and after how many jumps, do the chains starting from a l and a l +1 collide. This precomputation takes O(n *log n). Note that after the two chains collide, we now have the collision position k , and the two chains continue from a k and a k +1 , so we have exactly the same situation as in the beginning.

Now we can consider a new type of jump, from a l to a k as described above, and do binary lifting on those big jumps! This time we need to precompute two things for each l and i : where do we end up after 2 i big jumps, and how many small jumps does that entail.

And finally to correctly answer an [l , r] query, we first do the binary lifting steps using big jumps from a l , and as soon as the next big jump would overshoot a r , we do the binary lifting steps using small jumps for the remainder.

The solution was ultimately about applying a standard algorithm three times, not about coming up with a novel approach. However, I still find it quite beautiful how the same idea just keeps on giving.

Thanks for reading, and check back for last week's summary!

Thinking about growth and profit

James Long (jlongster)

I posted a follow-up twitter thread with a final spreadsheet and learnings.

In my last post I showed my process for forecasting costs and profits for Actual based on various metrics. This helps me find much-needed answers to questions like “how much cost for acquiring users can I afford?”

I’m new to this and I learned a lot from my first attempt. There’s still a lot to improve on though. This post goes into a few more things I’ve learned based on feedback and more research.

Some context: I launched Actual a year ago, but really only started focusing on growth since last September (4 months ago). It’s a very early product and I’m a solo founder.

First, a couple takeaways from responses to my previous post:

Let’s be more pessimistic about churn and say 10%.

Costs are simple in this model, but you want to all costs. I focus on Plaid costs because that will be the majority of my costs, but in a real forecast make sure to include everything like Stripe’s fees. Marketing should be part of acquisition costs, but I’m doing any marketing right now.

A reader asked an interesting question: how many people signed up for a trial without ever running the app (and never costing anything)? I can’t find the original response, but they said to expect almost half of trials to never do anything. I ran the numbers and so far on my 3 months of data, and:

Holy crap . Out of 1025 total trials, only 508 actually ran the app and logged in which is 49%. This is a little skewed because I’m transitioning away from a free plan which doesn’t track users, so some of those earlier trial conversions aren’t tracked. But it’s probably not far off.

For simplicity, I’ll assume all trial signups cost the same.

The cost of growth

Near the end of my last post, I asked the question “What does it take to hit $10k/month?” Cranking up growth surprised me because I found myself spending more and more money each month, spending $11,714 after 16 months with a 40% growth rate (link to spreadsheet):

It shouldn’t have surprised me though. Growth costs money, and simply put: in this scenario, I’m spending more than I’m making. The problem is I’m assuming a 40% growth for all time and given the cost per trial, conversion rate, and pricing I can’t sustain that.

All this money spent is really an investment . I may have spent a total of $42,746.16, but now I have 7834 subscribers. Let’s guess at a customer lifetime of 12 months, at $7/months I will eventually make 7834 * 7 * 12 or $658,056. That’s a 1539% return on investment.

In reality, constant 40% growth is not going to happen. Eventually growth will slow and I will start making a return. If it doesn’t slow, that’s amazing and you’re onto something big. Here’s what happens if I had 40% growth for 6 months, and then it slows down to 5%:

As growth slows, the costs I pay for trial users goes down and profits go up. If you wanted to, you could model growth as a curve over time and see how that effects cash flow. That would reflect reality since growth isn’t constant.

If you wanted to focus on growth for a long time, you’d either need to take a loan/investment or tweak other parameters (like pricing, conversion, etc) to support it.

Annual plans are killer

I briefly looked at annual plans at the end of the last post. With an annual plan, the user pays for an entire year upfront. Obviously, this provides a lot more cash flow.

It’s pretty incredible what this extra cash allows. Check out the (extremely hypothetical) 40% growth for 16 months (link to spreadsheet):

I never even lose money — I make $1012 the first month subscribers are charged. Sure, some of this money needs to be reserved to cover the operational costs of the user for a year, but it’s most likely a small fraction of that.

In a reply to my previous post, one person mentioned that if you offer monthly and annual plans, it’s common for 1/3 of your users to be on the annual plan. I’ll leave the modeling of this mixture up to you, but it’s not hard to guess that even with only a 1/3 you’d have a lot more freedom with cash flow.

Annual plans are killer.

Just how is growth and profit related?

Even with the annual plan, if you crank up growth you’ll still end up in the negative after 16 months. It may take a growth rate of 398%, but it still happens!

The math nerd in me had to figure out just how growth and profits are related. Obviously there is an inflection point where growth overtakes costs, but how do we solve for that? You don’t really need to understand all this , but at least try to understand why I had to redefine growth in the next section and then skip to the simpler math .

It would be nice to know, given acquisition cost, conversion rate, and pricing, just how much growth can I afford? A better way to put it: how much growth will I get if I reinvest all profits?

It took me a while to work through it. The first thing I did was add the ratio of total subscribers / new subscribers .

Sam Shuster: Predatory pirate journals and what we can do about them

julietwalker · The BMJ Blog

Once upon a time, the word “pirate” would recall the excitement of Treasure Island , but today, the Jolly Roger flag of warning is raised by publishers, whose pieces of eight come from academic predation. Predatory pirate journals are about quantity of money, not quality of purpose.

As online publishing developed, many academic journals allowed open access, costs being paid by the authors—in practice, by the universities and grant-giving bodies—and quality was maintained by editorial control and peer review. But financial predators saw this as a way to make money, since many people will pay to be published, and we are now being exposed to material that is dubious or incorrect. The number of predatory journals has increased since Beall’s original listing (Jeffrey Beall is an American librarian and library scientist, who first coined the term “predatory journals”), and more are using hijacked material. [1] This became apparent to me from the increasing number of emails received from journals I’d never heard of. The most revealing were invitations, sticky with flattery – “Dear Professor , I found your profile had a dynamic potential which fascinates me to email you.” Three or four of these now arrive daily, pleading for submission of a piece in any form and on any subject I choose; I am to be an instant expert in any branch of medicine or surgery. The commercial giveaway is that whatever I write must be submitted in just a few days, because one single article is needed to complete an issue for which fees have already been collected.

These absurd invitations provided me with the perfect opportunity to explore profit and editorial control. So, tongue in cheek, I replied to some that, having been invited, I presumed they would pay me a writer’s fee, and how much would it be? The reply was there was a misunderstanding, and I was expected to pay $2000 for publication, but a discount would be given. Haggling increased the discount even further. I tested other invitations by replying that as the piece was needed to complete an issue I would consider writing it if the fee was waived; a full or partial reduction was usually agreed.

Clearly, then, the primary concern of pirate journals is financial; but what about quality? All the journals I examined presented an impressive format, claiming that all submissions would be considered by independent reviewers. But the short time to publication made this unlikely, so to test this, I replied to invitations “ I can consider putting things aside to do the article only if you can guarantee its publication.” A typical reply was: “ We will guarantee for its publication”—so much for independent reviewing.

When payment is the criterion, quality comes second, which is why we are being increasingly exposed to the academic branch of the fake news family, with its low quality, un-refereed submissions. But a new development in predatory publishing was revealed when I came across an article falsely purporting to have been written by me. The publisher has not responded to requests to have this forgery taken down, and neither the postal address nor phone number given by the publisher are real. It seems that predatory journals now write their own “submissions” when needed. We must deal with this before it spreads further, with the added risk of undermining genuine academic publishing — but how?

We cannot stop the practice; that will happen only if and when our culture loses the primacy of money; but we can reduce its malign effect by increasing awareness of the problem and improving its detection — most pirate journals are new, and have a poorly edited, banal content, but they are becoming more sophisticated and less easily recognised. Awareness is important for people new to journal publishing and grant-giving and government bodies that award funding on the basis of publication quantity rather than quality; but this will need care so that it does not add to the atmosphere of disbelief that fake news has already provoked.

Finally, and tedious though it may be, the decisive tool can only be certification of academic reliability. To achieve this, academic publishers will have to collaborate with universities, professional societies and industry, to set up a body such as a Council for Academic Publication, to review and certify a journal’s acceptability. Approved journals could carry the emblem of approval, and an electronic reference confirming that material viewed or downloaded is from an approved journal.

Exposure and isolation of predatory pirate journals is urgent; it will need funds and a concerted effort to minimise the problem; such an approach could even make some pirates walk the plank to oblivion.

Sam Shuster , emeritus professor of dermatology, Woodbridge, UK.

Competing interests : none declared.

1 Macháček V, Srholec 2021, M Predatory publishing in Scopus: evidence on cross-country differences. Scientometrics https://doi.org/10.1007/s11192-020-03852-4

The post Sam Shuster: Predatory pirate journals and what we can do about them appeared first on The BMJ .

Folie à Machine: LLMs and Epistemic Capture

DaystarEld · LessWrong

"Truly, whoever can make you believe absurdities can make you commit atrocities." — Voltaire, 1765

A man in his late forties discovers a new passion. After decades of working as a mid level manager for a restaurant chain, he starts spending his evenings deep in study, filling notebooks with diagrams, reading everything he can get his hands on. Within a few weeks he's convinced he's on the verge of a grand unified theory that reconciles quantum mechanics and general relativity. He starts emailing professors at universities, posting on forums, and brushing off every criticism, rejection, or dismissal of his findings. His wife tries to have a conversation about it; he tells her she "just wouldn't understand." He's considering quitting his job to do it full time.

Would you consider this delusional?

A software developer has an idea for a startup. Not just any idea, the idea, the one that's going to change everything. He starts iterating on it, refining his pitch, building prototypes. His friends point out some problems with the concept: the unit economics don't work, there's no evidence of market demand, his core assumptions about user behavior seem unfounded. He listens politely, then explains why it's worth going forward anyway. He iterates again. And again. Each iteration is more elaborate, more detailed. He's been "about to launch" for six months. He's working sixteen-hour days and has never been more certain of what he's doing with his life.

Would you consider this delusional?

A fifty-year-old woman starts an online relationship with someone she's never met in person. They talk every day, sometimes for hours. He's charming, attentive, says all the right things. Unfortunately he lives overseas, can't video chat due to his work, has some financial difficulties she helps with. Her daughter Googles the photos he sent and finds them attached to a completely different person's social media, but when confronted, the woman has an explanation for that. And for every other red flag. She's sent thousands of dollars over the course of a year. She gets angry when people tell her it isn't real. She knows it's real. She talks to him every day.

Would you consider this delusional?

I expect most people would hesitate for at least one of these. We might say "obsessive," or "overconfident." We might say they need help. We might even say they've been brainwashed.

But "delusional" is a strong word, and psychosis even more so. These aren't people who hear voices or believe the CIA has implanted chips in their teeth. They're functional. They go to work (mostly). They can carry on normal conversations (mostly). They have reasons for what they believe, and if you sat down with them, they could articulate those reasons with apparent coherence.

And yet something has clearly gone wrong. Some mechanisms that should allow their models of reality to self-correct toward actual reality have been disrupted, and the longer it goes on, the harder it seems to be for them to come back. The usual feedback channels aren't working.

None of the examples above required an LLM. The amateur physicist could have gotten there through pop-science books and Reddit. The startup founder through hustle-culture seminars. The catfish victim through Facebook.

Still. A growing number of people have been noticing that LLMs are capable of inducing all of these states, and more, in a wide variety of people. People with no history of mental illness, and no apparent propensity to give up their default sensemaking apparatus to someone, or something, else.

Pathology vs Pathologizing

The phrases "LLM psychosis" or "AI psychosis" have been bouncing around online with increasing frequency. It isn't a clinical term yet, and the use of the word "psychosis" raises some people's hackles.

New technology always causes panics, and this might be one of them. The most reasonable version of this pushback I've seen lately is this post by DeepFates. "LLM Psychosis" doesn't seem to be referring to any coherent thing, but rather a bunch of things being lumped together under a bad label. And it's not only failing to cut reality at its joints, it's potentially flattening our ability to look around and notice what else may be happening when someone has an unusual experience with LLMs, what new sorts of interactions may be possible, that isn't easily categorized as "dysfunctional."

And as for the bad stuff, surely some of it is people having the mental health crises they would have had anyway, just now doing it in conversation with an AI, right? Others might be noticing real things about the world that happen to actually just be weird and new, or exploring bizarre but meaningful questions about the nature of these models and their inner experiences, or their understanding of humanity, or love, or the universe.

These are fair points, and under the umbrella of a concern I take pretty seriously. In my Philosophy of Therapy , I wrote:

"Pathologizing" is the perception that any action or view that is unusual is automatically a sign of illness, despite no evident dysfunction or suffering. In decades past, previous versions of the Diagnostic and Statistics Manual labeled things like homosexuality a mental health illness due to a mentality that didn't distinguish between "normal" and "healthy." Newer versions of the DSM have eliminated most of those, and there's a concerted effort among (good) psychologists and therapists to distinguish real pathology as something that causes direct suffering for the patient.

A polar week

Petr Mitrichev · Petr Mitrichev (competitive programming)

ICPC World Finals 2025 #ICPCBaku was the main event of the week, and for many the main event of the year (problems , results , top 12 on the left, official stream , our stream). The St Petersburg State University team was always among the leaders, but they collected too many incorrect attempts and therefore the almost flawless University of Tokyo team was leading with a better penalty time on 10 problems just two minutes before the end. However, the St Petersburg State University has managed to get the very difficult problem G accepted as well, and the university became the world champion for the 5th time . Congratulations!

For the University of Tokyo this means that they have earned their 13th medal and 6th gold , but will unfortunately need to wait to lift the cup for a bit longer. I expect them to do it soon, given the flourishing programming contest community in the university and in Tokyo in general. In any case, well done to them and to all medalists!

We have solved the mirror contest together with Kevin and Mateusz. It started reasonably well, but then initally Mateusz and then myself got stuck in problem B , spending a lot of thinking time and computer time on it, but never getting it accepted.

At the end of the contest, we tried to add some heuristics to speed up our matching code in B, but it was still too slow. The way that many teams used to work around it is to have a separate solution for “large enough” values of n , either a direct mathematical one or one obtained by looking for patterns in the outputs in the matching solution for small values of n .

However (it is always somehow easier to think after the contest ends...) today I have tried another approach that seems obvious in retrospect: since we just need to find an even number that is not covered by some matching, and in the normal bipartite matching algorithm (usually called Kuhn’s in the programming contest circles) as soon as we’re unable to find an augmenting path from a certain vertex of the first part, we know that it will not be covered by the final matching, we can stop the matching algorithm as soon as a search starting from an even number is unsuccessful. On my laptop, it makes the code run in 0.6s for $n=10^7$. I could not find any working upsolving to test this properly, please share a link if you know one!

Another problem that I found very nice was problem E : you are given a graph with $2n$ vertices and m edges ($n<=2*10^5$, $m<=4*10^5$), the vertices are split into n pairs that we will call blocks , let us call one vertex in a block red and another blue . Every edge connects a red vertex with a blue vertex. The m edges are added to the initially empty graph one by one. After each edge is added, you need to print the number of pairs of blocks (out of $n*(n-1)/2$ possibilities) that are connected (there exists a path connecting one of the four possiblities: between the red or the blue vertex in the first block and the red or the blue vertex in the second block). If not for the complication with the blocks, it would be a simple union-find application, but can you see how to deal with the blocks?

Thanks for reading, and check back next week!

Readable Unit Tests: make tests you actually want to write (and read)

Criteo Tech · Criteo Engineering

Author: David Haubenstricker

You'll read this test five times more than you'll write it. Future-you will thank present-you for mocking it readable.

This article distills and expands the talk I gave internally at Criteo: Readable Unit Tests . It's aimed at developers who know unit testing but avoid it because tests feel hard to write, hard to maintain, and — let's be honest — often ugly. Let's fix that.

TL;DR

Readable tests are:

Behavior-focused (test what the code does , not its guts)

Resilient to refactoring (survive reasonable changes)

Self-contained and obvious (you can understand them at a glance)

Consistent (follow familiar patterns, so your brain's pattern-matcher lights up)

Eight concrete practices help you get there:

Divide tests into given / when / then

Use good test names with clear conditions

Keep everything in the test (avoid shared setup/teardown)

Generate test data for non-significant values

Use creation functions to hide noisy details (especially mocks)

Prefer test data classes when parameters get complex

Consider scenario classes for readable end-to-end setup

Get feedback — what's readable to you today may not be tomorrow

What's a "unit" anyway?

A unit test targets a small, isolated piece of behavior — typically a method, class, or rule. There are two characteristics that all unit test must have that guide how "large" a unit we can handle.

Unit tests (all tests for that matter) need to be reliable . Ideally eliminating external factors that are out of our control. This generally means no external services, and nothing that could be holding state. Unit tests can interact with shared state like in-memory databases to test sql, if the tests are constructed well. In-memory databases are fast, easy to set up, and teardown between tests. It's best when it is possible to have multiple instances so tests are more isolated from each other.

Unit tests must also be fast , otherwise they won't be run frequently and they lose a lot of their value. Unit tests should be able to be run every code edit. Good guidance is that they should take 10 seconds or less.

There is then the question of how large a unit should be? Which generally boils down to a tension between clarity and resilience :

Bigger, API-level tests can be resilient to change (still work after refactoring code) but often obscure what's being exercised.

Smaller, focused tests are clearer but can be fragile if they poke at implementation details.

It is best to do a combination of both. Lots of smaller tests that are used to exhaustive verify that all code paths work as expected. Usually this requires a lot of mocking. The bigger tests validate that the interfaces between classes are working correctly. (We are mocking correctly.) They function like an internal integration test for your classes.

Strong opinions, lightly held

See the test fail once. Red → Green → Refactor. If you never saw red, you don't know the test works.

Layer your tests. Have requirement-level tests and focused unit tests.

Bugs hide in the dark code. Aim for thorough coverage, but remember: assertions determine test quality.

Code that's hard to test is bad code. Refactor it.

Great developers write great test code. Make tests a first-class citizen in your codebase.

The Readability Rules (with examples)

1) Divide tests into given / when / then

arrange / act / assert is another common naming convention. The goal is to use comments or blank lines to make structure obvious.

[Test] public void Check_ShouldRejectSku_WhenIsNotInStock() { /\/ given var context = GenContext(); var now = DateTime. UtcNow; var outOfStockSku = GenSku(context) with { Quantity = 0 }; var skuChecker = new SkuChecker(CreateConfiguration(context), CreateLineItemCache(now, outOfStockSku));

/\/ when var response = skuChecker. Check(context, now, outOfStockSku);

/\/ then Assert. That(response, Is. EqualTo(RejectReason. OutOfStock)); }

A thesis week

Petr Mitrichev · Petr Mitrichev (competitive programming)

Atto Round 1, also known as Codeforces Round 1041, was the main event of the week (problems , results , top 5 on the left, analysis). With problem H turning out way easier than the problemsetters have expected, the top places were decided on problem G concerning permutations and their compositions, which for me was just very hard to load into RAM to even think about it :) At the end of the contest I came up with a randomized solution that takes 15 random permutations and asks all their cyclic shifts. It actually passed G1 in upsolving, but during the round I did not manage to find all off-by-one and permutation-instead-of-inverse bugs. But even in upsolving it seemed hopeless for G2 where we only get to take 10 such permutations.

It turns out that Kevin has won the round with exactly the same solution. To pass G2, he tried 1000 different random sets of 10 permutations before starting to ask questions, and chose the set that gives the least collisions. This turned out to be borderline good enough, and 4 submissions that differed only in the random seed (diff on the right) sealed the deal. Well done Kevin, and also well done Jiangly on solving G2 in the “on paper” way :)

I got my share of successful randomized solving in problem H, where after being stuck for a while I decided to submit a solution that took edges that can definitely be taken, and when there is no such edge, took a more or less random edge and continued, and then repeated this process until the time runs out. This should be roughly the same as backtracking but even simpler to implement, and I expected that with n =30 it has a high change of passing, and indeed it did from the first attempt in which I fixed all bugs.

It is curious that neither problemsetters nor myself were able to see:

1) the relatively standard dynamic programming solution for the problem that takes advantage of the fact that from every subtree of the DFS tree we have at most 5 backward edges, and therefore can just add the state of all those edges to the dynamic programming state.

2) the non-bipartite matching solution that does not even rely on the additional “at most 5” property from the problem statement: the problem statement essentially asks for maximum matching, but where each vertex can be used twice instead of once, so we can just duplicate the vertices and be done.

Point 2 is even more bizarre in my case, since my masters thesis 18 years ago was on a more general version of exactly this problem, 2-path-packings. The thesis was in Russian and I don’t think it is published anywhere, but it is roughly equivalent to chapter 3 “O(mn)-time algorithm” in this paper . So the matching interpretation should really have been obvious to me, even after 18 years :)

Thanks for reading, and check back next week!

Jeffrey Aronson: When I Use a Word . . . Snowflakes

jross · The BMJ Blog

I love snowflakes. I enjoy crunching them underfoot on a crisp winter's day and the silky feeling that you get when skiing through a fresh fall.

The word "snowflake" entered English in the early 18 th century, first recorded in a pantomime called Cupid & Psyche, or, Colombine-Courtezan, based on the tale told by the Numidian writer Apuleius (c124–c170) in The Golden Ass: " Soft as the cygnet's down his wings, And as the falling snowflake fair". This was a surprisingly late entry, considering how early its separate components, "snow" and "flake", had appeared. The earliest instance of "snow" cited in the Oxford English Dictionary is in the Vespasian psalter , from c825 AD: " Se seleð snaw swe swe wulle ", which is an Old English translation of the Latin version of Psalm 147, verse 16: "qui dat nivem quasi lanam", or in the original Hebrew

which literally means "the giving of snow like wool", more clearly translated as "he spreads snow like a woollen blanket". "Flake" first appears, linked to snow, in Chaucer's House of Fame (c1384): "As flakes fallen in great snowes".

But words evolve and, like many other words, "snowflake" has accrued many other meanings.

Snow-buntings have white bodies and what appear to be flecks of snow on their wings. Starting in the late 17 th century, they were therefore called snow-flecks. Then, in the late 18 th century, that became corrupted into "snowflakes".

The snow-bunting, snow-fleck, or snowflake (Wikimedia commons).

At about the same time, William Curtis published his Flora Londinensis , which included the picture of a purple foxglove that William Withering used as the frontispiece to his 1785 monograph on digitalis, An Account of the Foxglove . Curtis gave the name snowflake to the Leucoium aestivum , to distinguish it from Galanthus species of snowdrops, both members of the family Amaryllidaceae . These plants contain galantamine , a cholinesterase inhibitor that has been used to treat dementias; the results have been unimpressive in both Alzheimer's disease and vascular dementias .

As in the psalm, snow and wool also come together in the use of the word "snowflake", when it describes a method of weaving woollen cloth.

A snowflake is also the name given to a hairline crack in metal, arising when the metal has not been allowed to cool down sufficiently slowly, and described in a 1919 issue of the Bulletin of the American Institute of Mining Engineers as a "white silvery area, which always has the appearance of being of a very coarsely crystalline structure".

Snowflakes appear in mathematics too, and I discuss one instance in the Interesting integer section below. Here is another. Draw an equilateral triangle (Figure a; black). Now on the middle third of each side draw another equilateral triangle and erase its base (b; red). Now repeat that on each available side in the new shape (c; blue). If you continue doing this you get a shape that resembles a snow flake (d). This is a fractal pattern, one that has the same local characteristics no matter how high a magnification you choose to look at it. The boundary is of infinite length, because you can go on adding new triangles forever. However the area it encloses is finite and is 60% larger than the area of the original triangle.

One more meaning of "snowflake" remains to be explored.

Ah, well, I should have expected that to happen

Finally, an aphorism from a book called Myśli nieuczesane nowe (More Unkempt Thoughts , 1964) by the Polish poet Stanisław Jerzy Lec: Żaden płatek śniegu nie czuje się odpowiedzialny za lawinę. No snowflake feels responsible for an avalanche.

Jeffrey Aronson is a clinical pharmacologist, working in the Centre for Evidence Based Medicine in Oxford's Nuffield Department of Primary Care Health Sciences. He is also president emeritus of the British Pharmacological Society.

Competing interests: none declared.

The post Jeffrey Aronson: When I Use a Word . . . Snowflakes appeared first on The BMJ .

EUC 2026 first day

Petr Mitrichev · Petr Mitrichev (competitive programming)

Yesterday I have arrived in Warsaw, which will host ICPC EUC 2026 this Sunday. As you can see on the left, Warsaw in Feburary (much like Wijk aan Zee in January) is very conductive for indoor activities, which is perfect to be able to truly focus on a programming competition.

I have arrived too late to get to the opening ceremony, but from the slides that were shared it seems that, among other things, Bill has announced the dates for the upcoming ICPC World Finals: November 15-20, 2026 in Dubai (which was already announced before), and September 14-19, 2027 in a yet undisclosed (but seemingly known to the ICPC organizers) location. The ICPC EUC 2027 location was announced last year to be Eindhoven, Netherlands.

But for now the focus is on the upcoming championship, where 8 teams have already qualified for the Dubai World Finals from the various ERCs and are just competing for fun and glory, while 8 more teams will qualify tomorrow. Best of luck!

Scripting News

I gotta say some days I start with a lot on my mind and am driven to write. This is one of those days. Maybe I'm inspired by the torrent of posts by my blogger friend ma.tt . Blogging can be a solitary thing or a relative thing. When you blog about something I have something to say about, I write on my blog and link back to yours, that's relative. The problem with comments in the old blogging world is that my comment resides on your blog. No more of that. I want equal stature for all writing, your comment should appear on your blog, yet still be easy to find from the other person's blog (and this is very important) with their support, it has to be something they want their readers to see. Otherwise the comment is still on your blog where your readers can see it.

Today's RSS cleanup

Scripting News

Did some work on my RSS feed this morning.

There was only one source:account element in my RSS feed and it was to a Twitter account I lost control of a few weeks ago, the account I had there since 2006, that at one time was in the top 10 accounts on Twitter. I changed the account to bullmancuso – one of my many testing sites on Twit, and added my Mastodon and Bluesky accounts. I will check the bullmancuso account every so often, but the masto and bluesky ones are several times daily. And the funny thing is that I have far more Bluesky and Mastodon accounts than Twitter accounts. Put that in your press release. ;-)

I updated the JSON version of the RSS feed, but noticed the Masto and Bluesky accounts (above) hadn't come through, and then on further investigation I realized that I hadn't been keeping it up to date for quite some time! The rule is every time something is added to the XML version I have to also add support for it in the JSON version. This is where Claude came in handy, it gave me a list of all the updates that were needed and I did them. We make a pretty good team imho. The two versions should be in sync now.

A note on my JSONification of RSS. When I first did it in 2010 , it was on a lark. I was tired of hearing how people didn't like XML so could I please switch. So I did and published it, and I think no one used it. The idea of a JSON version of RSS is in the air again, so I brought it up to date. I want to serve it with a different https-friendly address, but probably won't get to that today.

The reason I wanted to work on this today because I have to add RSS-generating code to wpIdentity , so I needed to get warmed up on that part of the world.

The new version of the daverss package is 0.6.15.

Scripting News

Thinking about linkblogging, my blogroll software doesn't do it correctly. When you click on the link to a linkblogged link, you must go to the place the linkblog entry points to, not the linkblog itself. I know that sounds confusing, but here's an example. It's obvious we can skip the stop and go right to the thing they were pointing to. It's awkward in the code because the RSS 2.0 item-level link element is doing double duty. I think I should add a source:linkblogLink element. I also think it's a good time to start discussing this among devs. There's some very nice fertile ground here and an opportunity to work with each other.

So I did and published it, and I think no one used it.

Scripting News

Scripting News

Quick note on Bluesky's disclosures . Yesterday they disclosed $100 million investment in April last year. It's good that they cleared it up, but bad that they were hiding it for so long. Everything about what they do is based on trust. New management probably is the reason this happened now. They should also clean up the promises they've made about Bluesky as a platform. I've done the homework, having developed a few apps using their API, some are still running. If I were their new CEO, I would announce that in addition to supporting AT Proto, they will also hook up Bluesky to the web. The web is already decentralized. Lots of developers know how to build web stuff. We can all breathe the same air.

A byte week

Petr Mitrichev · Petr Mitrichev (competitive programming)

There was no contest that I wanted to mention last week. Since I am getting such empty weeks quite often now, I've decided to check unfiltered clist in case I have missed something. And indeed, there seems to be an onsite round of a competition called Wincent DragonByte coming up in September which I only discovered today, and therefore missed my chance to qualify for the onsite finals in Bratislava :) It is a pity but in the grand scheme of things there are of course so many contests I already took part in and hopefully even more that I will take part in.

Nevertheless, this raises a logical question: what would be a good strategy to avoid missing such nice competitions? Reading all Codeforces posts does not seem to work, and neither does checking clist regularly enough. Is there a way to get notified when new contest websites are added to clist maybe (for example, maybe one can achieve that by excluding less relevant resources via "hide" rules in the filters, as opposed to choosing the relevant resources as "show", which I am doing now)? Or should I create an AI bot that checks unfiltered clist or unfiltered Codeforces and decides if I should be notified? What is your strategy? And of course, what other cool contests am I already missing? :)

Thanks for reading, and check back for this week's summary!

There was no contest that I wanted to mention last week.

Petr Mitrichev

Parkinson’s Law of Worry

Jakub Halmeš · LessWrong

Parkinson’s law says that “Work expands so as to fill the time available for its completion.” I think that a similar observation can be made for people worrying about stuff. To paraphrase: “ Problem salience expands so as to fill the capacity available for worrying.”

Suppose a person is worried about several problems. Let’s visualize the mental state of this person, where each problem is represented by a colored circle, and the size of the circle corresponds to how much this problem occupies the person:

This person worries about 5 things, with the yellow and green ones being the most important ones.

This person worries about 5 things, with the yellow and green ones being the most important ones.

Now, when one of these problems is resolved, one would expect that this problem simply gets removed from the ‘mental space’, making the person less worried in proportion to the size of the resolved problem:

The person now worries less, because the big problem was resolved.

Naive expectation: The person now worries less, because the big problem was resolved.

However, I don’t think this accurately describes what actually happens! Instead, it seems that the unresolved problems become more salient to the person, as if to fill the available space in the person’s mental state:

Or, new problems pop inside the freed space — the problems which weren’t important enough to worry about as long as there were more pressing ones:

The upshot is that in the beginning, the person would be wrong to think that they will worry much less or be much happier after the most important problems are resolved. They will just worry about different things!

What could help to prevent this scenario is to try to periodically recall some of the resolved problems and the amount of worrying they caused before they were resolved. By remembering this, the importance of other problems can shrink again (it’s relative, so they should still be less important than the yellow problem was!).

Related concepts: Adaptation level theory, hedonic treadmill, gratitude, negative visualization.

Scripting News

Thinking about the SAVE Act , 60 Minutes should do a segment on what you have to go through to get a birth certificate in any random state. It’s a lot of work, I’ve had to do it twice in the last few years. You’d have to be a pretty committed voter to be willing to do all that work. I imagine it would be even harder if you’re black, and it’s going to be hugely hard for married women who changed their last name when they got married. And how much you want to bet they don’t accept birth certificates from Muslim countries? It is the biggest scam ever, and if the journalists don’t cover it that way, always, with no both-sides-isms, then we should all know this is the end of journalism in the US. And btw also the end of real elections in the US too.

The Repubs these days like to say they’re against the “deep state” – well my friends this is about the most deep state bullshit ever.

Memory lane for Frontier users

Scripting News

I had to find out which domains being served by a problem server were still mapping to its domain. This server had been running for six years, and I was pretty sure some of the apps had moved.

So I wrote a script in Frontier, it was the best tool available to me, and got my answer in 20 minutes, code written from scratch.

The script visited each subfolder, the filename is the domain of the folder, finds out which server it's supposed to be running on, based on a DNS lookup, and adds a line to a list.

Here's a screen shot of the domains folder.

Here's the script as a screen shot and GitHub doc .

This is just a way to preserve a little of the Frontier culture. Hard to explain in words. Easier to show as screen shots.

ANALYSIS

IN BRIEF

Scripting News, Scripting News: When I heard about Matt's product Beeper I thought wow what if that were on the RSS network .

I think RSS should be here. Makes sense doesn't it?

Why not an open independent format from nowhere that no one objects to you using and will not do anything ever to turn you off. It seems it would be fairly easy to add two-way support. :-)

Scripting News, Scripting News: I asked Claude : "What is OpenClaw useful for? Do you think I could use it in my programming work, based on what you know about what I do?" Basically it's for non-programmers. Then I asked: "I wonder if I could make software that would be useful to people who love OpenClaw?" That was more interesting and included in the response I linked to, above.

Jill Lepore, The New Yorker: Jill Lepore writes about "The AI Doc: Or How I Became an Apocaloptimist" and A. I. constitutions, like that of Anthropic's chatbot Claude.

Scientific American, Scientific American: Decades of data have suggested the universe is flat, much like an infinite plane. But a new analysis reveals deep flaws in that simple conclusion

Scripting News, Scripting News: If I knew how AI would work with software, I would've done things differently to prepare for this. I find myself wanting to ask questions about my code that I don't have proper tools to answer. I have to get all my code managed with the new system, but not sure that's even the right way to go. Once I started using it to build full bits of deployed code, not to just answer questions about the work I'm doing one day at a time, I've become confused about planning my own work.

Scripting News, Scripting News: WordPress can now connect via MCP for both reading and writing. This sounds like a possible alternative for the wpcom api that we're building on in WordLand . Sometimes it feels like everything is being reinvented. If the world would just stand still for a moment we might be able to do some building. I wonder how the advent of AI is affecting how WordPress is being developed. I know it's changing everything here.

Scientific American, Scientific American: A Los Angeles trial jury found that Meta and YouTube are offering products that are addictive and harmful to young users' mental health

Hilton Als, The New Yorker: How the artists in this year's survey do or, more often, don't acknowledge those who paved the way for them.

Scripting News, Scripting News: The thing that we all missed is that WordPress is the best candidate for a standard for what an individual social network message is.

Scientific American, Scientific American: Atmospheric scientist Perry Samson was doing fieldwork when he was unexpectedly caught inside a tornado—making him one of the few such people who have lived to tell the tale

Scripting News, Scripting News: Podcast : A one-line comment on Brent Simmons' blog got me started on a 10-minute ramble about suspension of disbelief, in software. Also a story about meeting Ted Nelson at the West Coast Computer Faire in SF in 1979. Skiing. And other miscellanea. BTW, I didn't even remember the quote correctly and I might have misinterpreted it. It's still a good story imho. ;-)

Scripting News, Scripting News: If you're using FeedLand and running a WordPress blog, you can install a blogroll just like the one I have at scripting.com or blogroll.social .

Chris Wiley, The New Yorker: In a new exhibit, the Norwegian photographer finds divergent ways to break through and touch an audience numbed by visual glut.

Scientific American, Scientific American: Security feeds and traffic cameras have helped guide some of the most audacious targeted killings in modern history. Security researchers say the underlying vulnerabilities cover the planet and are easy to exploit

Scientific American, Scientific American: Skin conditions such as psoriasis often flare up in the same spots throughout one's life. Now scientists think they know why

Scientific American, Scientific American: Climate scientist Kate Marvel talked to Scientific American about her decision to leave NASA amid federal government turmoil and funding challenges

Scientific American, Scientific American: The science-fiction film Project Hail Mary sees Ryan Gosling go to space in a state of suspended animation. But does the science suggest that's possible?

Scientific American, Scientific American: In 2017 NASA's Hubble Space Telescope zoomed in on a comet as it passed around the sun. And then things took a more unusual turn

Patrick McKenzie (patio11), patio11 (Patrick McKenzie): I joined Stripe to work on building financial infrastructure for the Internet. Time flies. Here is what I've learned in the first four years.

The New Yorker, The New Yorker: The mega-popular K-pop stars have been on hiatus for nearly four years. Their new album, "Arirang," tests the group's staying power in the global cultural marketplace.

Scripting News, Scripting News: They've been having intelligent and clear-thinking guests on CNN and MSNOW on the coverage of the Iran War, unusually good discourse. But the best coverage I've heard has been from Frontiline podcasts. There's a new one out , haven't listened to it yet, but the one I heard yesterday was very informative and probably a better briefing than our president has been getting (or paying attention to).

Scientific American, Scientific American: A start-up has surprised the scientific community with a breakthrough: translating a modern proof into a programming language for verification using AI. But not everyone is celebrating

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